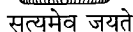


TECHNICAL PUBLICATION No. 23



2013

© NATIONAL CENTRE FOR ANTARCTIC & OCEAN RESEARCH, GOA

ISBN: 978-81-906526-7-4

Report Prepared by : Sh. L. Prem Kishore

Printed at : CSIR -- National Institute of Science Communication and
Information Resources, Dr. K.S. Krishnan Marg,
New Delhi-110 012

कपिल सिब्बल
KAPIL SIBAL



विज्ञान एवं प्रौद्योगिकी तथा महासागर विकास मंत्री
भारत सरकार, नई दिल्ली

MINISTER FOR SCIENCE & TECHNOLOGY
AND OCEAN DEVELOPMENT
GOVERNMENT OF INDIA
NEW DELHI

MESSAGE

I am delighted to send my warm greetings to the Wintering Team members at 'Maitri' on annual mid winter day on 21st June at Antarctica. The National Antarctic Programme owes its success to the sincerity and devotion displayed by you and your colleagues. Your relentless efforts in pursuit of scientific knowledge in the harsh conditions of Antarctica is keeping the image of India, bright and high in the International arena. We, the people of India admire your courage.

It gives me great pleasure in informing you that Government of India has recently given the approval for enhancing the Hard Duty Allowance by 50% over the existing rate. This is a small gesture from our side to acknowledge your dedication and sincere efforts in carrying out the mission objectives of our Antarctic Programme.

Best Wishes for a successful stay and safe return home

(KAPIL SIBAL)

Shri. L. Prem Kishore,
Station, Commander and
All the Members of the XXV Winter Team
'Maitri'
Antarctica



डॉ. प्रेम शंकर गोयल
DR. P. S. GOEL

सचिव
भारत सरकार
महासागर विकास विभाग
महासागर भवन, ब्लॉक-12, सी.जी.ओ. कॉम्प्लेक्स,
लोदी रोड, नई दिल्ली-110 003

SECRETARY
GOVERNMENT OF INDIA
DEPARTMENT OF OCEAN DEVELOPMENT
MAHASAGAR BHAVAN BLOCK-12 C.G.O. COMPLEX,
LODHI ROAD, NEW DELHI-110003

Date: 20th June 2006

MESSAGE

My most valued and dear friends,

Warm greetings to you all on the occasion of Antarctic Mid Winter Day on 21st June. Your courage and determination to work and explore in the most hostile conditions of Antarctica and keep the national flag flying is a matter of pride. We send you all our affectionate greetings and good wishes on behalf of numerous well wishers and friends in the Ministry and the Antarctic Scientific community. We look forward to your safe return after accomplishing the mission objectives, for which you have gone.

(P.S. Goel)

Shri. L. Prem Kishore,
Station Commander and
All the Members of the XXV Winter Team
'Maitri' Antarctica



डॉ. प्रेम शंकर गोयल
DR. P. S. GOEL

सचिव
भारत सरकार
पृथ्वी विज्ञान मंत्रालय
महासागर भवन, ब्लाक-12, सी.जी.ओ. कॉम्प्लेक्स,
लोदी रोड, नई दिल्ली-110 003

SECRETARY
GOVERNMENT OF INDIA
MINISTRY OF EARTH SCIENCES
MAHASAGAR BHAVAN BLOCK-12 C.G.O. COMPLEX,
LODHI ROAD, NEW DELHI-110 003
DO. No. MoES/Secy/8/2006
August 14, 2006

MESSAGE

Dear Dr. Prem Kishore,

Thanks for your letter. Let me first greet you and the entire team on the occasion of Independence Day. You are holding the Indian Scientific Spirits in adverse conditions of Antarctic. We respect the brave and committed scientists in the very cold and dark environment. I wish you all a very successful and exciting stay at Antarctic.

With kind regards,

Yours sincerely,

(P.S. Goel)

Shri. L. Prem Kishore,
Leader & Station Commander
XXV Indian Scientific Expedition
Indian Research Base Maitri
(70° 45' 59.8" S & 11° 48' (54.5")
ANTARCTICA



डॉ. शैलेश नायक
DR. SHAILESH NAYAK

सचिव
भारत सरकार
पृथ्वी विज्ञान मंत्रालय
पृथ्वी भवन, लोदी रोड, नई दिल्ली-110003
SECRETARY
GOVERNMENT OF INDIA
MINISTRY OF EARTH SCIENCES
PRITHVI BHAVAN, LODHI ROAD, NEW DELHI-110003

FOREWORD

The twenty fifth Indian Scientific Expedition to Antarctica was remarkable not only because it marked the two and a half decades of India's sustained presence in Antarctica, but also because of some spectacular successes the Team achieved on a scientific plane. Be it in retrieving a 64.5 m long ice core from the Humbolt Glacier, or carrying out bathymetric surveys of the Priyadarshini Lake for Human Physiology, or the microbial studies aimed at understanding the biotechnological potential of the credit. The observations made by the meteorologists and atmospheric scientists, the samples and the wealth of data collected by the geoscientists and biologists, and the results of the studies already carried out and in the pipeline are sure to go a long way in improving our understanding of the pristine continent, in general, and of the Schirmacher Oasis in particular.

Shri L. Premkishore from the National Geophysical Research Institute, Hyderabad, had the honour of leading India's Silver Jubilee Expedition to Antarctica. Ably supporting him in all his endeavors were the other 51 members of the Expedition, many of them veterans in their own right in Antarctica studies or logistics. The synergy between the Scientific and Logistics complement no doubt contributed to a very great extent in scripting the success story of the Expedition.

I am glad to note that the same enthusiasm has permeated to this excellent Technical Publication which summarises the initial results of the Expedition. My compliments to the entire team for so successfully accomplishing all the tasks entrusted to them.


(Shailesh Nayak)

राष्ट्रीय अंटार्कटिक एवं समुद्री अनुसंधान केन्द्र

पृथ्वी विज्ञान मंत्रालय
(भारत सरकार)
हेड लैण्ड सडा, वास्को-डा-गामा
गावा- ४०३ ८०४, भारत

Dr. S. RAJAN
Director



NATIONAL CENTRE FOR ANTARCTIC & OCEAN RESEARCH

Ministry of Earth Sciences
(Government of India)
Headland Sada, Vasco-da-Gama,
Goa - 403 804, INDIA.

PREFACE

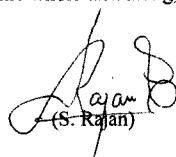
This Technical Publication synthesises the scientific and logistics accomplishments of the 52-members who constituted the team of the twenty-fifth Indian Scientific Expedition to Antarctica. Led by Shri L. Premkishore of the National Geophysical Research Institute, Hyderabad, the Team had 27 members who wintered over in Antarctica, while the rest returned to India after the austral summer of 2005-06.

Like in the previous years, the 25th Indian Scientific Expedition was also a story of all-round success, exciting in terms of the scientific output, innovativeness and team-spirit. A spectacular achievement was the retrieval of a 65.04 m long ice core from the Humbolt Glacier by the Geological Survey of India. The team carried out GPR surveys utilizing 100 and 200 MHz antennas which helped it in selecting the drill site. The GSI-NCAOR team carried out the bathymetric surveys of the Priyadarshini Lake using a portable single beam Echo sounder. Scientists from the Birbal Sahni Institute of Paleobotany carried out the mapping of paleolakes and collected samples from the dry lake beds for subsequent analytical studies.

On the Remote Sensing side, concurrent/non-concurrent ground truth data was collected over cap ice with respect to the IRS P6 satellite passes, using the indigenously developed radio meters. In the field of biological sciences, scientists from the Piramal Life Sciences collected soil samples for screening for their bioactivity and to isolate microbes. Botanical Survey of India on the other hand, undertook a floristic and taxonomic survey on the diatom communities of the freshwater lakes, pools and streams of Schirmacher Oasis. Another interesting study was initiated by DIPR on interpersonal relations among the expedition members during periods of isolation in Antarctica.

An unique feature of the Expedition was the introduction of a Student Participation programme for to inculcate interest in the younger generation on the study and significance of polar regions.

Shri Premkishore and his entire Team deserve all kudos not only for undertaking a successful Expedition but also for putting together in the most professional manner, their preliminary results in the form of this compilation. From NCAOR, a special word of Thanks to Shri Rasik Ravindra, former Director and Dr. Achuthankutty, Visiting Scientist for meticulously going through many versions of the draft which vastly improved the quality and substance of the different papers. Dr Rahul Mohan, as ever, pulled the whole task along, to the best of everyone's satisfaction, in his own inimitable way.


(S. Rajan)

CONTENTS

Page No.

MESSAGES

Minister for Science & Technology and Ocean Development,
Government of India iii

Secretary Ministry of Earth Sciences v

Secretary Ministry of Earth Sciences vii

FOREWORD ix

PREFACE xi

THE TEAM xix

DIARY OF EVENTS xxvii

Achievements during the Twenty Fifth Indian Antarctic Expedition 1-4

L. Prem Kishore

Leader's Report on XXV ISEA: An overview 5-34

L. Prem Kishore

EARTH SCIENCE, GLACIOLOGY AND GLOBAL CHANGE

Shallow Ice Core Drilling On Polar Continental Ice North of
Humboldt Glacier, East Antarctica 37-48

Arun Chaturvedi, A. Raj kumar, M. J. Beg, P. C. George

Temporal analysis of Meteorological data and the surface energy
balance from RRTS of the Blue Ice in Dronning Maud land in
East Antarctica 49-72

V. D. Mishra, Manoj Kumar and P. K. Srivastava

Climatic history and landscape evolution of Schirmacher Oasis,
East Antarctica during Late Quaternary: A multi-proxy study 73-105

Binita Phartiyal

A new approach for Bathymetric Survey in Antarctic Lake:
A Pilot study. 107-117

S. K. Chaturvedi

- Wide band magnetotelluric investigations to delineate the deep crustal geoelectric structure in East Antarctica 119-137
S. Prabhakar E. Rao, Sharana Basava and T. Harinarayana
- Studies on the increase in Seismicity in the Antarctic Plate: Observations from Broad Band Seismological Observatory (MAIT) at Maitri, Antarctica during XXV IAE 139-149
S. V. R. Ramachandra Rao, N. Ravi Kumar and E. C. Malaimani
- Geodetic tying of Antarctica and India for geodynamical and strain accumulation studies in the south of Indian Peninsula with Continuous GPS measurements at Maitri during 25th IAE 151-161
S. V. R. Ramachandra Rao, N. Ravi Kumar and E. C. Malaimani

ATMOSPHERIC SCIENCES

- Meteorological programme at Maitri, Antarctica during 25th expedition 165-177
P. A. V. Nampoothiri
- Measurements of Atmospheric Potential Gradient in Antarctica using Indigenously Developed Passive Antenna System 179-186
K Jeeva, K. U. Nair, Atul Kulkarni and Ajay Dhar
- Prospects of High-Frequency Acoustic Sounding at Maitri, Antarctica 187-194
H. N. Dutta B. S. Gera and N. C. Deb
- Space Weather and Irregular Ionospheric patches at "MAITRI" 195-202
Shailendra Saini and A. K. Gwal

BIOLOGICAL SCIENCES

- Freshwater Diatom communities of Schirmacher Oasis, East Antarctica 205-212
M. Palanisamy
- Ecological Study of Southern pole bird Skua and Penguins during Indian Antarctic scientific expedition to Antarctica. 213-222
Ashok Survwade

CONTENTS

xv

Page No.

Exploration of Microbial diversity from Antarctica with special emphasis on Biotechnological potential <i>Shailendra Sonawane, Girish Mahajan, Saji George, P. Giridharan, Asha Almeida, P. D. Mishra.</i>	223-234
---	---------

PHYSIOLOGICAL SCIENCES

Cognitive and Psychomotor Performance in Extreme and Unusual Environment <i>Dr. K. Ramachandran, Dr. F. U. John Paul, and Dr. Manas K. Mandal</i>	237-257
Interpersonal Relations among Indian Antarctic Expedition Members in Isolated and Confined Environment <i>Dr. K. Ramachandran, Dr. F. U. John Paul, and Dr. Manas K. Mandal</i>	259-276
Personality and Temperament of Indian Antarctic Expedition Members <i>Dr. F. U. John Paul and Dr. K. Ramachandran</i>	277-284

ENGINEERING AND LOGISTICS

Technical Report on Antarctic Communication <i>Narendra Singh & Ajay Sharma</i>	287-292
Medical Report of the XXV-ISEA <i>Dr. (Mrs) Devyani D Borole</i>	293-298
Logistic Report <i>Lt. Col. Janardhan Rao,</i>	299-319

ACKNOWLEDGMENT

At the outset in all humility and gratefulness I would like to thank Dr. P.S. Goel, the then Secretary, Department of Ocean Development and all the members of the expert committee for reposing the faith and giving me an opportunity to lead the most important and momentous 25th Indian Scientific Expedition to Antarctica – “A Silver Jubilee Expedition”.

I would like to thank Dr. H.K. Gupta former Secretary, Department of Ocean Development, for his wisdom, knowledge and commitment to the highest standards which inspired and motivated me for taking up such a big challenge.

I am greatly indebted and like to express my immense gratitude to Dr. P. C. Pandey the then Director NCAOR, for his whole hearted support, guidance and motivation that has encouraged me to take up the challenging task.

Thanks are also to Dr. V. P. Dimri, former director, National Geophysical Research Institute, Hyderabad for giving me an opportunity to participate and Lead the 25th ISEA.

Any attempt at any level in Polar Regions could not have been satisfactorily completed without the active support and guidance of Shri Rasik Ravindra, Director NCAOR. I am greatly indebted to him for his constant support, words of encouragement and motivation that has given me and the whole team and raised the morale of the team to come out successfully.

I also thank Dr. M Sudhakar Rao, Incharge Director, NCAOR for organizing and launching the expedition.

I am also greatly indebted to Shri Indra Mohan, Ex-Scientist, and NGRI as he was always a constant guide and mentor who have initiated me to step into the Icy Continent and for taking up big challenges, providing all the necessary guidance and support to me throughout.

I am also thankful to Shri T.V.P. Bhaskara Rao for being with me and the whole team throughout, right from the beginning of the expedition for his sustained support, guidance and for his coordinate effort and calm ability to resolve any sort of last-minute problem. Making all the necessary arrangements whenever it was required and giving all the inputs for the expedition and also to his support team

consisting of M/s Reena, Bhavani, Shradha and Rajashree for making all the necessary arrangements right from the beginning of the expedition.

I also like to express my immense gratitude and thanks to Dr. N. Khare for his coordinated efforts, guidance and valuable suggestions at appropriate times in guiding me throughout the expedition.

Thanks are also to Shri Arun Chaturvedi as an Observer of expedition whose co-operation was always there with me and for his support, guidance and making me feel comfortable at many a times.

Thanks are also due to Mr Javed Beg for his invaluable support and liaison to carry out many tasks and outstanding contribution in establishing the summer camp and organizing the field camps and successfully completing the summer programmes.

I will fail in my duty if I don't acknowledge with thanks the unstinted support of Col. R. Janardhana Rao and his team for looking after the logistics which are the core of success of any expedition.

The valuable contribution of Cmdr Gurumani is duly acknowledged and I thank him for taking responsibility and looking after the basic needs of the expedition members during the voyage.

Thanks are also due to Capt Kingston and his team for rendering valuable and innovative engineering services throughout the expedition.

I am also thankful to all the marvelously, excellent diverse participants, their good ideas and desire to build the spirit of the expedition who have rendered their whole hearted support at all the times for the successful completion of the expedition.

I also like to thank Dr. Rahul Mohan for guiding me and helping me in bringing out this Report.

I also want to thank my wife Vanitha and sons Jaynth Nag and Ananth Nag without whom, my own participation would have been difficult. Their love, support and continuing patience have taught me so much about sacrifice, discipline and compromise. They always encouraged and believed in me, in all my endeavors and who so lovingly and unselfishly with profound understanding made me what I am today.

L. Prem Kishore
(L. Prem Kishore)

THE TEAM

25th Indian Scientific Expedition to Antarctica

Summer Team:

S.No.	Name	Organisation
1.	Shri Arun Chaturvedi	Geological Survey of India (Observer)
2.	Shri Mervin J. D'souza	Geological Survey of India
3.	Shri Mirza Javed Beg	Geological Survey of India
4.	Shri P. C. George	Geological Survey of India
5.	Shri S. Prabhakar Eknath Rao	National Geophysical Research Institute
6.	Dr. Sharana Basava	National Geophysical Research Institute
7.	Shri Anil K. Mehta	Survey of India
8.	Shri Anurag K. Sharma	Survey of India
9.	Shri Atul Kulkarni	Indian Institute of Geomagnetism
10.	Dr. H. N. Dutta	National Physical Laboratory
11.	Dr. Subodh K. Chaturvedi	National Centre for Antarctic and Ocean Research
12.	Dr. A. Rajakumar	National Centre for Antarctic and Ocean Research
13.	Cdr. J. Gurumani	National Hydrographic Office, Indian Navy
14.	PO SRI, Yashpal Kalra	National Hydrographic Office, Indian Navy
15.	LSSR-I, Vinoj Kumar S	National Hydrographic Office, Indian Navy
16.	Dr. (Ms.) Binita Phartiyal	Birbal Sahani Institute of Palaeobotany
17.	Dr. M. Palanisamy	Botanical Survey of India

S. No.	Name	Organisation
18.	Dr. Vivek Mishra	Barkatulla University
19.	Shri Shailendra Saini	Barkatulla University
20.	Shri M. R. Panwar	Defense Institute of Psychological Research
21.	Ms Preetee Oswal	National Botanical Research Institute
22.	Shri Ashok G. Surwade	MES Abasaheb Garware College
23.	Shri Shailendra K. Sonawane	Nicholas Piramal Research Centre
24.	Shri Pinak Ranade	Centre for Development of Advanced Computing
25.	Shri Jagmal Singh	Defense Electronics and Application Laboratory
26.	Nk/Mason, Sukhdev Singh	Indian Army
27.	Ct/Ck. Ram Dass	Indo-Tibetan Border Police

THE TEAM

Summer Team Members of the 25th ISEA



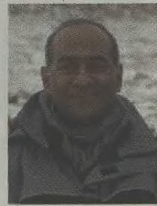
Shri Arun
Chaturvedi, GSI



Shri Mervin
J. D'souza, GSI



Shri Mirza
Javed Beg, GSI



Dr. H.N. Dutta
NPL



Shri P.C. George
GSI



Dr. M. Palanisamy
BSI



Dr. (Ms.) Binita
Phartiyal, BSIP



Dr. A. Rajakumar
NCAOR



Dr. S.K.
Chaturvedi
NCAOR



Shri S. Prabhakar
Eknath Rao
NGRI



Dr. Sharana
Basava
NGRI



Shri Atul
Kulkarni
HG



Dr. Vivek Mishra
BU



Shri Shailendra Saini
BU



Shri M. N. Panwar
DIPR

Summer Team Members of the 25th ISEA

Cdr. J. Gurumani
NHO, Indian Navy



PO, Sri, Yashpal Kalra
NHO, Indian Navy



LSSR-I, Vinod Kumar S.
NHO, Indian Navy



Shri Anil K. Mehta
SOI



Shri Anurag K. Sharma
SOI



Shri Pinak Ranade
CDAC



Shri S. K. Sonawane
NPRC



Shri Jagmal Singh
DEAL



Ms Preeti Oswal
NBRI



Shri Ashok G. Surwade
MES College



Nk/Mason, Sukhdev
Singh Engrs, IA



Ct/Ck. Ram Dass
ITBP

Winter Team:

S.No.	Name	Organisation
1.	Shri L. Prem Kishore (Leader)	National Geophysical Research Institute
		
2.	Shri S. V. R. Ramchandra Rao	National Geophysical Research Institute
3.	Shri Manoj Kumar	Snow and Avalanche Study Establishment
4.	Shri P. A. Vasudevan N	India Meteorological Department
5.	Shri S. Venkateshwara Rao	India Meteorological Department
6.	Shri K.U. Nair	Indian Inst. of Geomagnetism
7.	Shri F. U. John Paul	Defense Institute of Psychological Research
8.	Lt. Col. R. Janardhan Rao	EME, Indian Army
9.	Sub. B. K. Sharma	EME, Indian Army
10.	Nb/Sub. M. Palaniappan	EME, Indian Army
11.	Nb/Sub. E. J. Cutina	EME, Indian Army
12.	Nb/Sub M. G. Chavan	EME, Indian Army
13.	Nb/Sub H. C. Pandey	EME, Indian Army
14.	L/Nk Manoj Kumar	EME, Indian Army
15.	L/Nk. D. Chakravarthy	EME, Indian Army
16.	Capt. J. Kalyan Kingston Yesudas	Engrs. Indian Army
17.	Hav/OEM Pradeep Kumar V	Engrs. Indian Army
18.	Spr. Sanjjai Singh Bist	Engrs. Indian Army

S.No.	Name	Organisation
19.	L/Hav/C&J Ramesh Singh	Engrs. Indian Army
20.	Nk. Abraham Mathew	Engrs. Indian Army
21.	Mr. Narendra Singh	Defense Electronics Applications Laboratory (DEAL)
22.	Mr. Ajay Sharma	Defense Electronics Applications Laboratory (DEAL)
23.	Dr. Deviyani Borole	National Centre for Antarctic and Ocean Research
24.	Dr. Dhanjaya Patil	National Centre for Antarctic and Ocean Research
25.	Ct/Ck Vijay Kumar	Indo-Tibet Border Police

THE TEAM

Winter Team Members of the 25th ISEA



Shri L. Prem Kishore
Leader and Station
Commander, NGRI



Shri P.A. Vasudevan
IMD



Shri K.U. Nair
IIG



Shri S.V.R. Ramchandra Rao, NGRI



Shri S.Venkateshwara Rao, IMD



Shri Manoj Kumar
SASE



Shri F.U. John Paul
DIPR



Mr. Narendra Singh, DEAL



Mr. Ajay Sharma
DEAL



Dr. Deviyani Borole
NCAOR



Dr. Dhanjaya Patil
NCAOR



Lt. Col. R. Janardhan Rao, EME, IA



Capt. J. K. Kingston Y
Engrs. IA

Winter Team Members of the 25th ISEA

Sub. B.K. Sharma
EME, IA



Nb/Sub. M Palaniappan
EME, IA



Nb/Sub. E.J. Cutina
EME, IA



Nb/Sub M.G. Chavan
EME, IA



Nb/Sub H.C. Pandey
EME, IA



L/Nk Manoj Kumar
EME, IA



L/Nk. D. Chakravarthy
EME, IA



Hav/OEM Pradeep
Engrs, IA



Spr. Sanjjai Singh Bist
Engrs, IA



L/Hav/C&J Ramesh
Singh, Engrs, IA



Nk. Abraham Mathew
Engrs, IA



Ct/Ck Vijay Kumar
ITBP

Diary of Events

Chronology of Events, 25th Indian Expedition to Antarctica 2006-2007

L. Prem Kishore

Leader of 25th ISEA

16.12.2005	Welcome function to all Members of 25 th - ISEA at NCAOR, Goa.
17.12.2005	Fire Fighting and Orientation Lectures.
18.12.2005	Lectures by NCAOR Invitees.
19.12.2005	Expedition launching Function.
20.12.2005	Departure from Goa.
21.12.2005	Departure from Mumbai. Arrival at Cape Town, South Africa.
22.12.2005	Briefing by Dr P.S. Goel, Secretary, Department of Ocean Development at Indian High Commission, Cape Town.
24.12.2005	Embarkation on Expedition vessel.
29.12.2005	Sailed for Antarctica.
30.12.2005	Crossed 40° South Latitude.
4.1.2006	First ice berg sighted.
5.1.2006	Sighting of few whales.
8.1.2006	Crossed Antarctic Circle at 08:40 am. Celebrations on Heli.deck of M.V. Paardeberg.
9.1.2006	Encountered the Pack Ice and Penguins. The vessel reached at 20:00 hrs 69° 50' S 12° 27' E from where Helicopter operations could start for Maitri Station.
10.1.2006	Reached Antarctica. First courtesy Helicopter sortie to Maitri.
11.1.2006	Summer Camp established at Maitri with induction of 11 members along with equipment and cargo.
13.1.2006	15 more members inducted into summer camp along with equipment and cargo. Installation of AWS Tower for SASE.

- 15.8.2006 Independence Day celebrations with severe blizzard.
- 16.8.2006 Celebration of Krishna janmashtami Mega Serial Sri Krishna started.
- 24.8.2006 A major Overhaul of Alternator of Surya-I completed.
- 27.8.2006 Celebrations of Ganesh Chaturthi
- 5.9.2006 Celebrations of Onam festival with Lunch.
- 12.9.2006 Overhaul of 62.5 KVA Generator Aditya-IV completed.
- 20.9.2006 Overhaul of 62.5 KVA Generator Aditya-I completed.
- 1.10.2006 Celebrations of Dushera Festival with Russian Guests.
- 4.10.2006 Ninth convoy left Maitri at 09:00 AM with 6 Vehicles and 10 persons on Board.
- 7.10.2006 A Severe Blizzard within a span of just 18 hours huge accumulation of Snow. Convoy team held up at Shelf.
- 12.10.2006 Ninth Convoy returned back at 22:30 hrs to Maitri.
- 15.10.2006 Celebrations of 63rd Anniversary of EME Corps with a Salute of Snow Petrels the first visitors to Maitri.
- 17.10.2006 Tenth Convey left Maitri at 07:00 hrs with 4 Vehicles and 10 persons on board.
- 23.10.2006 Tenth convoy returned back at 19:00 am with two vehicles and 10 persons on board.
- 24.10.2006 First visitor of Maitri "Skua" arrived at 11:00 hrs.
- 26.10.2006 Preparation of summer camp with marking and layout of the sleeper. Four living modules placed.
- 29.10.2006 Remaining six living modules placed.
- 30.10.2006 Two bath modules installed.
- 31.10.2006 Five individual modules installed.
- 1.11.2006 Eleventh convoy left Maitri at 07:00 hrs to the shelf and then to the Russian barrier for bringing 40 Kl of fuel for the ALCI flight.
- 2.11.2006 The first ALCI flight arrived at Novo air strip at 03:30 AM and left on the same day at 15:30 PM.

14.1.2006	14 more members inducted into summer camp along with equipment and cargo.
15.1.2006	Summer camp fully established and field programs of BU, Nicholas, NBRI, ZSI, CDAC commenced. One Reece sortie made to Gruber Mountains with GSI and M.T. Group of NGRI. Equipment dropped at the first point of observation.
16.1.2006	Mr Arun Chaturvedi made a reconnaissance sortie to the drilling site.
26.1.2006	Republic day celebrations with Russians guests. A cricket match between 24 th and 25 th ISEA teams, 25 th team won the match.
29.1.2006	First convoy to Ice Core drilling site.
30.1.2006	Return of first convoy from ice core drilling site at 02:30 hrs.
9.2.2006	Reconnaissance sortie for convoy route.
15.2.2006	Second Convoy to drilling site with 4 vehicles and 7 persons on board.
16.2.2006	Return of convoy from drilling camp at 01:00 hrs.
20.2.2006	Induction of 24 th -ISEA members back onto the ship
21.2.2006	Handing and taking over of the station. Decanting of fuel started on the shelf.
24.2.2006	Third (convoy) with 8 vehicles and 15 persons on board started from Maitri at 09:00 hrs to shelf.
1.3.2006	Return of the convoy with 9 vehicles and 9 persons on board at 04:00 hrs.
3.3.2006	The ship "M.V. Paardeberg" sailed off to Cape Town for return voyage with 24 th ISEA members.
7.3.2006	The Fourth convoy started at 06:30 hrs along with 7 vehicles and 10 persons on board.
9.3.2006	The Fourth convoy returned back at 07:00 hrs
16.3.2006	30 KVA generator Jeevan Jyothi overhauled.
17.3.2006	Fifth convoy started at 07:30 hrs with 8 vehicles and 10 persons on board.
20 th March	Overhaul of 62.5 KVA generator Aditya-IV completed.

21 st March	Fifth convoy returned back at 21:30 hrs with 8 vehicles and 9 persons.
3 rd April	Sixth convoy started at 07:30 hrs with 8 vehicles and 9 persons.
6 th April 2006	Sixth convoy returned back at 23:30 hrs with 8 vehicles and 9 persons on board.
11 th April 2006	Seventh convoy started around 07:00 hrs with 6 vehicles and 10 persons on board.
14 th April 2006	Seventh convoy returned back at 23:30 hrs with 7 vehicles and 10 persons on board.
21 st April 2006	Eighth convoy started at 07:00 hrs with 5 vehicles and 10 persons on board.
24 th April 2006	Eighth convoy returned back at 03:00 am with 5 vehicles and 10 persons on board.
27 th April 2006	Preparation of Demands list started.
15 th May 2006	Modification of containers for storage of Generator spares completed.
20 th May 2006	Overhaul of 62.5 KVA generators Suya-I completed.
25 th May 2006	Preparation and arrangements for a new Gymnasium and Sauna taken up with painting.
29 th May 2006	Inauguration of Gymnasium and Sauna by the Russian leader Mr Nikoloy.
30 th May 2006	Increase of Hard Duty Allowance, message from Director, NCAOR.
1 st June 2006	Mega Serial Ramayana started along with Yoga classes in the morning.
5 th June 2006	Maitri winter sporting activity started with the inauguration of Caroms Singles.
21.6.2006	Mid Winter Day celebrations, First Blizzard of the season.
26.6.2006	Mega Serial Ramayana ends. (Total of 26 Cassettes)
29.6.2006	Mega Serial Mahabharata started.
19.7.2006	First Sunrise after Polar Nights.
09.8.2006	Overhaul of alternator of Aditya-III Completed.
13.8.2006	Mega Serial Mahabharata ends. (Total of 47 Cassettes)

- 4.11.2006 The second visitors “Penguins” at Maitri.
- 8.11.2006 Summer camp established with the new living and bath modules.
- 9.11.2006 The first batch of the 26th ISEA along with the Mr Marc, Engineer from Kashbohrer arrived by ALCI flight at Novo Airstrip at 03:30 AM and were received with warm welcome at Maitri station at 08:00 AM. Settled the team in the new summer camp.
- 10.11.2006 Eleventh convoy returned back to Maitri at 11:00 PM along with 5 vehicles and 6 members on board.
- 11.11.2006 Welcome party to the first batch of the 26th ISEA and Mr Marc.
- 17.11.2006 Twelfth convoy left Maitri at 09:00 AM along with the Ice Core Drilling Party to the Shelf.
- 18.11.2006 Celebrations of 26th Anniversary of Corps of Engineers, Flag Hoisting and Cricket Match.
Visit of Seven Members from Japan (JARE) along with a Russian.
Twelfth convoy returned back to Maitri at 11:00 PM after establishing the Ice Core Drilling Site.
- 20.11.2006 The 125 KVA Bhaskara generators commissioned after a successful engine overhaul and modification of the cooling systems, a huge and successful exercise after working on it over a period of three months
- 21.11.2006 Sighted two Penguins with eggs on way to Novo.
- 30.11.2006 The second batch of the 26th ISEA arrived by ALCI at 03:00 AM and reached Maitri at 08:00 Hrs.
- 1.12.2006 9 members from Germany and 5 members from Norway along with a Russian visited Maitri.
Aerial visit to Vostok station for Para dropping of fuel barrels by ALCI.
- 2.12.2006 Thirteenth convoy left Maitri at 12:00 PM.
- 3.12.2006 Visit of 7 guests from UK, Norway, SA.

- 4.12.2006 Thirteenth convoy returned back to Maitri at 11:00 hrs from shelf along with the Ice Core drilling party after selecting a new site.
- 6.12.2006 Received the Cargo from ALCI.
- 8.12.2006 8 Crew members of IL-76 along with Novo team visited Maitri.
- 25.12.2006 Celebrations of Christmas with guests from Novo.
- 31.12.2006 New Year eve celebrations at Novo.
- 1.1.2007 New Year celebrations at Maitri.
- 14.1.2007 Celebrations of Sankranti.
- 15.1.2007 Celebrations of Army Day.
- 26.1.2007 Celebrations of Republic Day with Russian leader and members of Novo.
- 11.2.2007 Farewell function by 26th ISEA to the 25th ISEA.
- 13.2.2007 Arrival of the Dignitaries Dr. P.S. Goel, Shri Rasik Ravindra and Shri Prakash Kumar.
- 14.2.2007 Visit to Novo Station along with the Dignitaries.
- 15.2.2007 Inauguration of Pump House by Dr P.S. Goel.
First Helicopter sortie by the 26th ISEA.
Return Journey of the Dignitaries along with 4 members of 25th ISEA by ALCI from Novo Air field.
- 17.2.2007 Handing over the station to the new leader of the 26th ISEA.
Return journey of 19 members of the 25th ISEA by ALCI from the Novo Air field.
- 18.2.2007 Arrival at Cape Town 08:30 hrs and checked into the Hotel.
- 19.2.2007 Stay at Cape Town.
- 22.2.2007 Started from Cape Town and reached Johannesburg. Boarded the flight at Johannesburg for onward journey to Mumbai.
- 23.2.2007 Reached Mumbai All the 19 members left to their respective destinations.